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United States Patent
Kind Code
Date of Patent
Inventor(s)

12415247
B2
September 16, 2025
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Disc tool

Abstract

A disc tool for a processing machine for grinding and/or polishing a workpiece (W); wherein the disc tool (**10A, 10B, 10B**) has a carrier (**20**) for attachment to an output (**85**) of the processing machine (**80**); wherein a cushion element (**40A, 40B, 40C**) made of elastic material and a retaining wall element (**41A, 41B, 41C, 41D**) for retaining the cushion element (**40A, 40B, 40C, 61**) are arranged on the carrier (**20**); wherein the retaining wall element (**41A, 41B, 41C, 41D**) is held on the carrier (**20**) by means of retaining bolts (**35**); wherein retaining bolt passage openings (**48**) of the retaining wall element (**41A, 41B, 41C, 41D**) extending through retaining mounts (**28**), of the carrier (**20**) are accommodated along bolt sections (**36**) of the retaining bolts (**35**) extending along the bolt longitudinal axes (BL), and the retaining wall element (**41A, 41B, 41C, 41D**) is held between bolt heads (**37**) of the retaining bolts (**35**) and the carrier (**20**); and wherein an adhesive layer (**50**) is arranged on a side of the retaining wall element (**41A, 41B, 41C, 41D**) opposite the carrier (**20**), on which adhesive layer a processing means (**70**) for grinding and/or polishing the workpiece (W) can be releasably attached to the disc tool (**10A, 10B, 10B**). The adhesive layer (**50**) has recesses (**52**) assigned to the bolt heads (**37**).

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Appl. No.: 17/890388

Filed: August 18, 2022

Prior Publication Data

Document Identifier
US 20230059593 A1

Publication Date
Feb. 23, 2023

Foreign Application Priority Data

DE	10 2021 121 566.8	Aug. 19, 2021
EP	22184758	Jul. 13, 2022

Publication Classification

Int. Cl.: B24B23/02 (20060101); B24B7/18 (20060101); B24D9/08 (20060101)

U.S. Cl.:

CPC B24B23/028 (20130101); B24D9/08 (20130101); B24B7/186 (20130101); B24D9/085 (20130101)

Field of Classification Search

CPC: B24B (23/028); B24B (23/02); B24B (7/186); B24B (7/18); B24B (7/182); B24B (41/002); B24B (37/12); B24B (37/14); B24B (37/22); B24D (3/002); B24D (3/00); B24D (3/001); B24D (3/004); B24D (3/007); B24D (3/02); B24D (3/34); B24D (9/085); B24D (9/08); B24D (9/00); B24D (13/20); B24D (18/0045); B24D (2201/00); B24D (7/00); B24D (7/04); B24D (7/10); B24D (7/18); B24D (7/16)

USPC: 451/329

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Background/Summary**BACKGROUND OF THE INVENTION**

(1) The invention relates to a disc tool for a processing machine, in particular a grinding machine or polishing machine, for grinding and/or polishing a workpiece; wherein the disc tool has a carrier for attachment to an output of the processing machine; wherein a cushion element made of elastic material, in particular foam, and a retaining wall element for retaining the cushion element are arranged on the carrier; wherein the retaining wall element and in particular the cushion element are retained on the carrier by means of retaining bolts; wherein retaining bolt passage openings of the retaining wall element extending through retaining mounts, in particular screw mounts, of the carrier are accommodated along bolt sections of the retaining bolts extending along the bolt longitudinal axes, and the retaining wall element is held, in particular tensioned or clamped, between bolt heads of the retaining bolts and the carrier; and wherein an adhesive layer is arranged on a side of the retaining wall element opposite the carrier, on which adhesive layer a processing means for grinding and/or polishing the workpiece, in particular an abrasive means or polishing means, can be releasably attached to the disc tool.

(2) Such a disc tool is described in patent application DE 10 2020 105 572, for example. In this disc tool, the retaining wall element has the adhesive layer, namely a VELCRO™ (hook and loop fastener) layer, to which an abrasive sheet, for example, can be fastened. The adhesive layer is uneven in the region of the bolt heads, with which the retaining wall element and the cushion element are attached to the carrier, which is rigid compared to the cushion element and retaining wall element, whereby, for example, there is increased wear of the processing means or abrasive sheet, and/or a processing quality of a workpiece processed with the disc tool suffers. The disc tool itself, in particular the adhesive layer thereof, is also subjected to greater load during operation and wears out more quickly. The problem occurs to a particular degree with a processing machine in the form of a grinding machine that drives the grinding disc to make eccentric or oscillating grinding movements, in which high shear forces or transverse loads occur transversely to the output axis of the processing machine. Furthermore, the problem is exacerbated when the grinding disc has larger diameters.

SUMMARY OF THE INVENTION

(3) It is therefore the object of the present invention to provide an improved disc tool.

(4) In order to achieve this object, a disc tool of the type mentioned at the outset provides for the adhesive layer to have recesses assigned to the bolt heads.

(5) The recesses are advantageously present before the retaining bolts are screwed into the carrier.

(6) A cross-section or mounting cross-section of the recesses for mounting the bolt heads is preferably larger than a cross-section or insertion cross-section of the retaining bolt passage openings provided for inserting the bolt sections of the retaining bolts.

(7) The recesses advantageously extend around the retaining bolt passage openings, in particular radially around the retaining bolt passage openings with respect to the bolt longitudinal axes.

(8) A method for producing a disc tool, which has the features in the preamble of claim 1 or the features mentioned above, provides for the following: producing the recesses assigned to the bolt heads in the adhesive layer; arranging the retaining wall element on the carrier after the recesses are produced; fastening the retaining wall element to the carrier by inserting the retaining bolts into the retaining mounts, wherein the bolt heads are arranged in the recesses; and fixing the retaining wall element between the bolt heads of the retaining bolts and the carrier, in particular tensioning the retaining wall element or clamping the retaining wall element between the bolt heads and the carrier.

(9) The recesses advantageously extend around the retaining bolt passage openings, in particular radially around the retaining bolt passage openings with respect to the bolt longitudinal axes.

(10) A cross-section or mounting cross-section of the recesses for mounting the bolt heads is preferably larger than a cross-section or insertion cross-section of the retaining bolt passage openings provided for inserting the bolt sections of the retaining bolts.

(11) It is preferred if the cushion element is also attached to the carrier by arranging the retaining wall element on the carrier or in connection with the arrangement thereof on the carrier.

Accordingly, the following is advantageous: arranging the cushion element with the retaining wall element on the carrier after the recesses are produced; fastening the cushion element on the carrier by inserting the retaining bolts into the retaining mounts, wherein the bolt heads are arranged in the recesses.

(12) When the cushion element is sandwiched between the retaining wall element and the carrier, the following step is useful: tensioning the cushion element between the bolt heads of the retaining bolts and the carrier.

(13) The retaining wall element is used to hold the cushion element, which can also be characterized as at least one cushion element. In any case, the at least one cushion element is arranged between the adhesive layer and the carrier.

(14) Advantageously, the invention also provides for a processing machine, namely a grinding machine or a polishing machine, which has a disc tool of this type, wherein the processing machine has a drive for the disc tool, by means of which the disc tool can be driven to make a rotary and/or eccentric and/or oscillatory movement by the processing machine.

(15) A basic concept of the present invention is that there are recesses in the adhesive layer in the region of the bolt heads. For example, this makes it possible for the bolt heads to displace the elastic material of the cushion element and/or the retaining wall element in the region of the recesses, for example by means of a corresponding compressive load on the retaining wall element. However, the ambient pressure that occurs, for example when dust is extracted through the disc tool and/or due to pressure exerted by the operator of the disc tool, has an effect on the adhesive layer due to the recesses such that it does not have any bulges, for example, or has bulges to a lesser extent, or has uneven surface sections.

(16) The retaining wall element comprises, for example, a fabric, in particular a textile fabric, a film, or also composite materials, which comprises a combination of the aforementioned components, for example a retaining wall element which comprises a film and a fabric layer that are connected to one another, for example glued to one another.

(17) The retaining wall element is advantageously flexible and/or more flexible than the carrier.

(18) The disc tool carrier consists of a rigid material such as hard plastic, metal, or the like. The carrier can have rib structures, for example. In any case, the carrier is the preferably completely or essentially rigid component of the disc tool, while the retaining wall element and the at least one cushion element are flexible and adaptable compared to the carrier.

(19) The bolt heads act on the retaining wall element. Due to the recesses on or in the adhesive layer, the bolt heads do not impede the mobility of the adhesive layer during operation of the

processing machine or to a lesser extent than when there are no recesses. The adhesive layer as a whole can have floating mobility with respect to the carrier, for example, without the bolt heads forming imperfections or anchor points, so to speak, which would impede the mobility of the adhesive layer.

(20) Retaining bolt passage openings are provided on the retaining wall element and advantageously on the cushion element for the bolt sections.

(21) The bolt heads project laterally in front of the bolt sections, i.e. they have larger diameters or cross-sections than the bolt sections. A diameter or cross-section of the recesses is larger than a diameter or cross-section of a respective retaining bolt passage opening provided for inserting a bolt section of a retaining bolt.

(22) The recesses comprise, for example, cutouts and/or depressions in the adhesive layer. The recesses are already present before the retaining bolts are attached to the adhesive layer, for example in that the adhesive layer and/or an adhesive layer element having the adhesive layer is provided with the cutouts or depressions before the retaining bolts are introduced and/or attached, for example by pressing, thermal deformation, machining, or the like.

(23) It is advantageously provided that a central axis or a center of at least one recess is concentric with respect to the bolt longitudinal axis of the retaining bolt accommodated in the recess. However, it is also possible for the bolt longitudinal axis to be eccentric with respect to the central axis or the center of the recess.

(24) It is preferred if at least one of the recesses or all recesses of the adhesive layer has or have an inner diameter that is at least as large as an outer diameter of a bolt head accommodated in the recess or larger than an outer diameter of a bolt head accommodated in the recess.

(25) Furthermore, it is possible for a retaining bolt passage opening to be arranged in the interior of one of the recesses or all recesses of the adhesive layer, through which passage opening a respective bolt section of the bolt accommodated in the recess penetrates or can penetrate. Consequently, the recess extends around the retaining bolt passage opening, in particular in a ring shape.

(26) The bolt sections of the retaining bolts pass through retaining bolt passage openings in the adhesive layer. The recesses extend around the retaining bolt passage openings. The retaining bolt passage openings can have a smaller inner cross-sectional area than the recesses. However, it is also possible for the recesses to have an inner cross-sectional area in which a respective bolt head can be completely accommodated, so that the recess simultaneously forms a retaining bolt passage opening for the bolt section of a retaining bolt.

(27) The adhesive layer can have passage openings, in particular to provide the recesses, through which the bolt heads can penetrate the adhesive layer. Although the adhesive layer has an inner cross-section in the region of the recesses that is smaller than an external cross-section of the bolt head accommodated in the recess, it is still possible for the bolt head to slide past the adhesive layer during assembly or when screwed into the carrier, so that the bolt head is not supported on the adhesive layer in the fully assembled state and/or in the state in which it is retained on the carrier in the retaining wall element. Then the adhesive layer can even protrude in the direction of the bolt head.

(28) It is also advantageous if at least one of the recesses or all recesses in the adhesive layer has or have a larger cross-section and/or diameter than the bolt heads, so that a space is advantageously available between an inner circumference of a respective recess and an outer circumference of the bolt head accommodated in the recess transverse to the bolt longitudinal axis thereof.

(29) The adhesive layer preferably has less material thickness in the regions thereof which are arranged, in particular tensioned or compressed, between a respective bolt head and the carrier than in a region next to the respective bolt head. A material thickness of the adhesive layer is therefore preferably less in those regions that are arranged between bolt heads and carriers than in the other regions of the adhesive layer.

- (30) The adhesive layer can be arranged directly on the retaining wall element, so that the retaining wall element is processed by the aforementioned processing, so to speak.
- (31) However, it is also possible for the adhesive layer to be arranged on an adhesive layer element, which in turn is arranged on the retaining wall element. In this case, the adhesive layer element is provided with the recesses, for example with cutouts and/or depressions, before the retaining bolts are introduced. If the adhesive layer element is multi-layered and/or comprises several layers, the recesses assigned to the retaining bolts preferably extend through all layers of the adhesive layer element. It is possible for the recess to have different geometries, for example different cross-sectional areas, cross-sectional geometries, and/or cross-sectional surface areas, and/or the like, on at least two layers or plies of the adhesive layer element. Thus, a respective recess can have a larger cross-section or a larger cross-sectional surface area in the region of the adhesive layer than in a section that is arranged closer to the retaining wall element. It is advantageous if the recesses of the adhesive layer element are designed as passage openings with a passage cross-section through which the bolt heads can penetrate.
- (32) The adhesive layer element, together with the cushion element and the retaining wall element and optionally further layers or walls arranged on the adhesive layer element, in particular between the adhesive layer element and the carrier, preferably forms a cushion or pad, which can be fastened as a whole to the carrier using the retaining bolts and used to hold the processing means.
- (33) The disc tool has a processing side, on which the adhesive layer is arranged, and a machine side opposite the processing side, on which the disc tool can be connected to the processing machine. A tool mount is arranged there, for example.
- (34) An advantageous measure provides for a tool mount or a tool-mounting element having a tool mount to be arranged on the carrier, wherein the disc tool can be releasably connected to an output of the processing machine using the tool mount.
- (35) The cushion element consists, for example, of foam, rubber, or the like. The cushion element is designed, for example, as a so-called pad. It is possible for the cushion element to consist of different materials, for example layers of different foam materials.
- (36) At this point, it should be mentioned that the disc tool can also have at least two cushion elements, of which a first cushion element is tensioned or can be tensioned between the bolt heads and the carrier, while at least one second cushion element is arranged on a side of the retaining wall element facing away from the carrier. The at least one second cushion element is positioned closer to the adhesive layer and farther from the carrier than the first cushion element. The at least one second cushion element is arranged between the adhesive layer and the retaining wall element.
- (37) The cushion elements can be made of different materials, for example different foams. It is also possible that one cushion element consists of rubber or caoutchouc while another cushion element consists of foam.
- (38) The cushion elements are preferably superimposed in the manner of layers between the carrier and the adhesive layer. More than two cushion elements can easily be present. In this case, a cushion element, i.e. a first cushion element, is preferably arranged between the carrier and the retaining wall element or forms a part of the retaining wall element, while at least one second cushion element or several second cushion elements is or are arranged on a side of the retaining wall element facing away from the carrier.
- (39) The at least one second cushion element or further cushion elements preferably has or have passage openings for the bolt heads such that the bolt heads can penetrate this second cushion element without tensioning it against the carrier.
- (40) It is possible for at least one cushion element to be arranged on the side of the retaining wall element facing the carrier and/or on the side of the retaining wall element opposite the carrier or facing away from the carrier.
- (41) A single cushion element can thus be arranged between the retaining wall element and the carrier, for example the above-mentioned first cushion element. Furthermore, the retaining wall

element may be interposed between a single cushion element and the carrier.

(42) However, it is also possible for at least one cushion element to be arranged on respective opposite sides of the retaining wall element, i.e. for at least one cushion element to be present between the carrier and the retaining wall element and one between the retaining wall element and the adhesive layer.

(43) A cushion element arranged between the carrier and the retaining wall element, referred to above as the first cushion element, is sometimes also referred to as a sandwich cushion element in the following description. A cushion element which is arranged on the retaining wall element on a side of the retaining wall element opposite the carrier is also referred to as an adhesive layer cushion element in the following description. This cushion element is, for example, the aforementioned second cushion element. However, if both variants of cushion elements or both possible arrangements of cushion elements with respect to the retaining wall element are meant, the term cushion element is generally used.

(44) It is advantageous if the cushion element or one of the cushion elements is held on the carrier by the retaining wall element, so to speak.

(45) A preferred embodiment provides that the cushion element is arranged as an intermediate layer between the carrier and the retaining wall element.

(46) The bolt sections of the retaining bolts pass through the retaining wall element and the cushion element, for example.

(47) It is advantageous in this case if the cushion element and/or the retaining wall element is compressed between the bolt heads of the retaining bolts and the carrier. In any case, in this embodiment, the cushion element can be compressed by the bolt heads. Due to the displacement of the cushion element and/or the retaining wall element by the bolt heads, it is possible for components of the cushion element and/or the retaining wall element to protrude laterally in front of the bolt heads, so to speak.

(48) It is advantageously provided that the retaining wall element consists of a material with a higher tensile strength than the cushion element.

(49) The retaining wall element consists, for example, of a textile fabric, a nonwoven, a VELCRO™ (hook and loop fastener) layer, or the like. The retaining wall element tensions the cushion element against the carrier, so to speak. The retaining wall element is loaded by the bolt heads in the sense of a compression of the cushion element, so that material of the cushion element located next to the bolt heads is also subjected to a force in the direction of the carrier.

(50) In a preferred embodiment, the retaining bolts are screw bolts which can be screwed into the retaining mounts of the carrier, which are embodied as screw mounts. However, bayonet contours, for example, on the retaining mounts and the bolt sections, which can be brought into engagement with one another, are also possible. However, the retaining bolts can also be designed, for example, as rivet elements, tie rods, or similar components with which the retaining wall element and optionally the cushion element can be tensioned in the direction of the carrier. Furthermore, the retaining bolts can, for example, have clamping contours and/or gripping contours, for example Christmas tree contours, on their bolt sections for clamping engagement and/or gripping engagement from behind with the retaining mounts.

(51) It is preferred if the cushion element and the retaining wall element can be detachably fastened to the carrier. For example, the retaining bolts can be actuated using an assembly tool, for example a screwdriver, and accordingly have the actuating contours, in particular on the bolt heads thereof. Thus, if necessary, for example when it is worn out, the cushion element can easily be dismantled from the carrier and another cushion element that is not worn can be mounted.

(52) The carrier is preferably plate-like or disc-like. The carrier preferably supports the cushion element over the entire surface thereof. In principle, however, it is possible for the cushion element to protrude over an outer circumference of the carrier or to be set back behind an outer circumference of the carrier.

- (53) It is also advantageous if the retaining wall element completely covers the cushion element on its side facing the adhesive layer when the cushion element is arranged between the retaining wall element and the carrier. When the retaining wall element is arranged between the cushion element and the adhesive layer, it advantageously completely covers the carrier-facing side of the cushion element.
- (54) The disc tool preferably has a circular outer circumference or a circular outer circumferential contour. However, the disc tool can also easily have an elliptical or other non-round outer circumferential contour. Furthermore, the disc tool can have a polygonal outer circumferential contour, for example a triangular or rectangular outer circumferential contour.
- (55) It is advantageously provided that the carrier and/or the cushion element and/or the sandwich cushion element and/or the retaining wall element and/or the components to be explained below, for example an intermediate wall element and/or an adhesive layer cushion element and/or an adhesive layer element which has the adhesive layer, have identical or essentially the same outer circumferential contours.
- (56) The disc tool is advantageously provided with at least one flow-passage channel for air, which has inflow openings on the adhesive layer and outflow openings on the carrier, wherein air can flow through the at least one flow-passage channel from the adhesive layer in the direction of the carrier or, vice versa, from the carrier in the direction of the adhesive layer. Thus, for example, air laden with dust or particles can be extracted through the disc tool. The at least one flow-passage channel or flow-passage channels preferably penetrates or penetrate all of the layers and/or plies of the disc tool. Outflow openings are advantageously arranged on the machine side of the disc tool, for example on the side of the carrier facing away from the processing side. It is preferred if a number of flow-passage channels are arranged in a ring around the tool mount.
- (57) It is possible for the bolt head or bolt heads to be flush with the adhesive layer. It is preferred if the bolt head or bolt heads do not protrude in front of the adhesive layer in the direction of the processing means. It is also advantageous if the bolt head or bolt heads are arranged behind a plane into which the adhesive layer extends. Thus, in the plane of the adhesive layer, there are regions, in particular around the bolt heads, at which the processing means is not or cannot be supported on the adhesive layer and/or on the bolt heads.
- (58) Advantageously, the disc tool provides that a component of the retaining wall element and/or the cushion element displaced by the bolt head does not protrude in front of the adhesive layer, for example is aligned flush with the adhesive layer or is advantageously arranged behind the adhesive layer.
- (59) It is preferred if at least one of the recesses or all recesses in the adhesive layer has or have a larger cross-section and/or a larger diameter than the bolt heads, so that a space is available between an inner circumference of a respective recess and an outer circumference of the bolt head accommodated in the recess, transverse to the bolt longitudinal axis thereof. Thus, for example, there is an annular space, transverse to the bolt longitudinal axis, around the respective bolt head between the outer circumference thereof and the inner circumference of the recess.
- (60) It is advantageously provided for at least one of the recesses or all recesses to form a displacement cavity into which components of the retaining wall element and/or the cushion element displaced by a respective bolt head can be displaced or are displaced.
- (61) A preferred concept provides that at least one of the recesses or all recesses is or are formed and/or designed to accommodate a bulge of the cushion element and/or the retaining wall element formed by the respective bolt head. For example, if a respective bolt head displaces the cushion element or retaining wall element in such a way, in particular due to direct loading of the cushion element or the retaining wall element or also due to loading of the retaining wall element acting on the cushion element, that a kind of bulge or similar arch is formed in the direction of the adhesive layer, this bulge or arch can be accommodated in the displacement cavity. The adhesive layer thus forms a contact surface for the processing means, for example the abrasive means, which is

essentially flat and/or does not have any arching produced by a material displaced by a respective bolt head and protruding in the direction of the processing means.

(62) In the region of the recess, the adhesive layer can be completely interrupted or omitted. Consequently, the adhesive layer can have a hole or a passage opening, so to speak, in the region of the recess.

(63) It is also possible that the adhesive layer has only a reduced material thickness or height in the region of the recess, i.e. that the recess is a depression in the adhesive layer or comprises a depression in the adhesive layer.

(64) It is possible for a respective recess to have a single depth with respect to the adhesive layer. For example, the recess can have a flat bottom for accommodating the bolt head, and at least one of the recesses or all recesses is or are deeper in a region radially closer to the bolt longitudinal axis of the retaining bolt accommodated therein with respect to the adhesive layer than in a region further away radially from the bolt longitudinal axis.

(65) One embodiment can provide that the cushion element is supported directly on the carrier.

(66) A connecting wall element is advantageously arranged between the cushion element and the carrier. The connecting wall element preferably forms a thinner layer than the cushion element. The connecting wall element can easily be another, in particular thin, cushion element.

(67) It is possible that the connecting wall element only serves as a kind of buffer and cannot withstand higher mechanical loads than, for example, the cushion element. However, it is preferred if the connecting wall element consists of a material with a higher tensile strength than the elastic material of the cushion element.

(68) The connecting wall element and/or the retaining wall element and/or the adhesive layer wall, which will be explained below, and/or the intermediate wall element, which will also be explained, consist, for example, of a nonwoven material, velour, a textile fabric, a film material, a polishing means material, a VELCRO™ (hook and loop fastener) material, or the like.

(69) It is advantageous if the adhesive layer completely or essentially completely covers a side of the disc tool facing away from the carrier, wherein, however, openings can be present, namely, for example, flow-passage openings for air, in particular for extracting dust and/or an opening is provided for fastening the disc tool on an output of the processing machine, for example for a screw with which the disc tool can be screwed to the processing machine.

(70) The adhesive layer preferably covers at least 80%, preferably at least 85% or 90%, of the side of the disc tool facing away from the carrier.

(71) It is preferably provided that the cushion element and the retaining wall element and/or the cushion element and the connecting wall element form a sandwich-like connected unit or a composite element. The recesses are preferably provided or produced on this unit or the composite element, for example by the thermal and/or machining and/or abrasive and/or cutting and/or embossing process explained below.

(72) Furthermore, this composite element also has the flow-passage openings or flow-passage channels for air, for example for extracted air laden with particles and/or for air that can flow out of the disc tool on the processing side of the disc tool when it is operated with the processing machine. The advantage can be seen in the fact that the composite element can be inserted into a punching tool, for example, in order to punch the flow-passage openings or flow-passage channels. In this way or with this punching tool, retaining bolt passage openings provided for the retaining bolts can also be punched. Thus, the passage openings for the retaining bolts and the flow-passage openings or flow-passage channels can be produced in a single machining process. The disc tool can be held in a tool mount in which it can optionally remain in order to carry out subsequent processing steps, for example the processing steps explained below. However, these processing steps can also be carried out if a further tool mount is used for holding the disc tool or one of the components thereof.

(73) An advantageous measure provides that the recesses are formed by a thermal and/or machining

and/or abrasive and/or cutting and/or pressure-loading and/or embossing process that reduces a material thickness of the adhesive layer and/or an adhesive layer element having the adhesive layer. (74) For example, it is provided that hooks present on the adhesive layer, in particular VELCRO™ hooks, are plastically deformed and/or melted by thermal processing, so that they do not protrude as far in the direction of the processing side of the disc tool facing away from the carrier. However, it is also possible to use a grinding tool or a machining tool, for example a milling tool, to process the adhesive layer or the adhesive layer element in order to form the recess. Finally, it is also possible for the recess to be cut out or punched out partially or as a whole from the adhesive layer, for example using a punching tool or other similar cutting tool.

(75) In the case of the disc tool, it is advantageously provided that it has an adhesive layer element which has an adhesive layer wall with the adhesive layer and is arranged on the retaining wall element.

(76) The adhesive layer element can have several plies or layers, as will become clearer below. However, it is also possible for the adhesive layer element, so to speak, to comprise only a single ply or layer, for example having only the adhesive layer. Consequently, the adhesive layer element forms, for example, an additional single or multi-ply layer which is present on a side of the cushion element facing away from the carrier, in addition to the retaining wall element.

(77) It is advantageously provided with the disc tool that the adhesive layer element is held on the retaining wall element in a material and/or form-fitting manner. For example, it is possible for the retaining wall element to have a hook arrangement and/or VELCRO™ hooks for holding the adhesive layer element. Additional bonding creates a material connection between the retaining wall element and the adhesive layer element.

(78) An advantageous measure provides that the adhesive layer element has an adhesive layer cushion element made of an elastic material, in particular foam, arranged between the adhesive layer wall and the retaining wall element.

(79) It is possible for the cushion element, as already discussed, to be arranged between the adhesive layer wall and the retaining wall element. This cushion element can be the only cushion element of the disc tool and form the adhesive layer cushion element in this case.

(80) However, the disc tool can also have the already mentioned sandwich cushion element and additionally the adhesive layer cushion element. Further cushion elements are easily possible.

(81) It is advantageous if the adhesive layer cushion element has passage openings through which the bolt heads pass. It is advantageous if the bolt heads can be freely inserted or are inserted through these passage openings.

(82) It is also advantageous if the adhesive layer cushion element is not tensioned against the carrier by the bolt heads and/or is not loaded, in particular directly loaded, by the bolt heads.

(83) An advantageous concept provides that the adhesive layer cushion element is arranged on the adhesive layer element on a side facing away from the bolt heads.

(84) The cushion elements of the disc tool, for example the adhesive layer cushion element and the previously discussed cushion element located between the carrier and the retaining wall element, can consist of different or the same materials, for example rubber, foam, or the like.

(85) Advantageously, the disc tool provides for the adhesive layer element to have an intermediate wall element which is arranged between the adhesive layer cushion element and the retaining wall element and is connected to the retaining wall element, in particular is materially connected. The intermediate wall element is preferably made of a material that has a higher tensile strength than the adhesive layer cushion element, for example a textile material, nonwoven material, velour, or the like. The intermediate wall element and the connecting wall element can consist of the same material.

(86) It is preferred if the intermediate wall element and the retaining wall element are held together in a form-fitting and/or material-fitting and/or force-fitting and/or surface-fitting manner. For example, the intermediate wall element and the retaining wall element can be connected to one

another using a VELCRO™ (hook and loop fastener) connection. Gluing or another similar material connection, which is provided as an alternative or in addition to the form-fitting connection or VELCRO™ (hook and loop fastener) connection, is also readily possible for connecting the intermediate wall element and retaining wall element.

(87) An advantageous measure provides that the adhesive layer cushion element and the intermediate wall element and/or the adhesive layer wall form a sandwich-like connected unit or an adhesive layer composite element on which the recesses are formed. The same measures that can be provided for the composite element already mentioned, which has the cushion element arranged closer to the carrier, can also be provided for the adhesive layer composite element, for example flow-passage openings or flow-passage channels for air, for example for extracted air laden with particles and/or for air which can flow out of the disc tool on the processing side of the disc tool when it is operated with the processing machine.

(88) It is possible for the recesses to comprise passage openings on the adhesive layer or to be formed thereby, it being possible for the bolt heads to thereby completely penetrate the adhesive layer. It is therefore advantageous if the bolt heads are not supported on the adhesive layer.

(89) The passage openings have, for example, an insertion cross-section that is larger than a cross-section of the bolt heads, so that the bolt heads can easily penetrate the passage openings.

(90) However, it is also possible for the passage openings to have a smaller diameter than the bolt heads and for slots on the adhesive layer to extend radially outwards from the passage openings. Thus, the material of the adhesive layer can be displaced from the bolt heads when the bolt heads penetrate the adhesive layer.

(91) It can be provided that the retaining wall element directly forms the adhesive layer element.

(92) In the case of the disc tool, it is advantageously provided that the adhesive layer is arranged on the retaining wall element and/or the retaining wall element integrally comprises the adhesive layer.

(93) A preferred concept provides that the adhesive layer has a VELCRO™ (hook and loop fastener) layer with VELCRO™ hooks for holding the processing means.

(94) The adhesive layer and the cushion element advantageously consist of materials different from one another or are different layers. For example, the cushion element consists of foam or rubber, while the adhesive layer is preferably formed by a separate body that has a different material than the cushion element, i.e., it does not consist of foam and rubber, for example.

(95) The retaining wall element advantageously has retaining bolt passage openings for the bolt sections of the retaining bolts, from which passage openings slots extend radially away, so that a region of the retaining wall element surrounding a respective retaining bolt passage opening is segmented. The slots can be produced, for example cut or punched, for example in connection with the production of the flow-passage openings or flow-passage channels and/or the retaining bolt passage openings. It is particularly preferred if the slots are produced on the composite element having the retaining wall element and the cushion element.

(96) In the case of the disc tool, it is advantageously provided that the processing means is an abrasive sheet. It is also possible that the processing means is or comprises, for example, a knitted fabric, a nonwoven, a polishing fabric, or the like.

(97) The disc tool can comprise the processing means, for example the abrasive sheet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the invention are explained below with reference to the drawing. The following is shown in the figures:

(2) FIG. 1 a schematic view of a processing machine and a sectional view of a disc tool arranged on the processing machine approximately along a section line A-A; which is shown in

- (3) FIG. 2 which shows a cushion element and an adhesive layer element of the disc tool according to FIG. 1 in a perspective view from the processing side thereof;
- (4) FIG. 3 a section B of FIG. 1;
- (5) FIG. 4 a schematic view of the processing machine according to FIG. 1 and a sectional view of a second disc tool, approximately along a section line C-C; which is shown in
- (6) FIG. 5 which shows a cushion element and an adhesive layer of the disc tool according to FIG. 4 in a perspective view from the processing side thereof;
- (7) FIG. 6 a section D of FIG. 4;
- (8) FIG. 7 a sectional view of a third disc tool that can be operated with the processing machine according to FIGS. 1 and 4, approximately along a section line E-E; which is shown in
- (9) FIG. 8 which shows a cushion element with an adhesive layer the disc tool according to FIG. 7 in a perspective view from the processing side thereof;
- (10) FIG. 9 a section F of FIG. 7;
- (11) FIG. 10 an exploded view of the disc tool according to FIGS. 1-3 and the sub-components thereof;
- (12) FIG. 11 a section of an adhesive layer of a disc tool with signs of wear, approximately corresponding to a section G in FIG. 8;
- (13) FIG. 12 a partial section of a disc tool having the adhesive layer according to FIG. 11, approximately corresponding to the section according to FIG. 6, wherein the adhesive layer is unprocessed;
- (14) FIG. 13 a partial section of the disc tool according to FIG. 6, approximately corresponding to section D, wherein VELCRO™ hooks of the adhesive layer are shown;
- (15) FIG. 14 a schematic view of processing of a composite element for a disc tool using a thermal processing tool;
- (16) FIG. 15 a schematic view of processing of a composite element for a disc tool using an abrasive or cutting processing tool; and
- (17) FIG. 16 a variant of the exemplary embodiment of FIGS. 1-3, approximately corresponding to a section as in FIG. 3.

DETAILED DESCRIPTION

- (18) A processing machine **80** is, for example, a grinding machine and/or a polishing machine. The processing machine **80** has, for example, a housing **81** from which a handle **82** protrudes, which can be gripped by an operator.
- (19) The handle **82** can be a short handle.
- (20) Preferably, the handle **82** is designed as an elongated handle bar, on one end region of which a handle for grasping by an operator is arranged and on the other end region of which the housing **81** is arranged, so that a ceiling or a wall, for example, can be processed. For example, the processing machine **80** is a so-called long-neck grinder.
- (21) An electric and/or pneumatic drive motor **83** is arranged in the housing **81** and drives an output **85** via a gear **84**, for example in a rotary and/or eccentric and/or oscillating manner. Corresponding oscillating and/or eccentric and/or rotary driving gears **84** are known. The output **85** forms a tool mount and is arranged in an extraction housing **86**.
- (22) Disc tools **10A**, **10B**, and **10C** explained below are optimally adapted to operation with the processing machine **80** because they can withstand the forces and loads generated by the output **85** that are necessary, for example, for grinding or polishing a workpiece **W** using the disc tools **10A**, **10B**, **10C**.
- (23) Insofar as the disc tools **10A**, **10B**, **10C** have the same or similar components, they are provided with the same reference symbols in the following description. Wherever possible, similar components are provided with the same numbers, which, however, are provided with additional letters A, B, and C for assignment to the respective disc tool **10A**, **10B**, **10C**. When all disc tools **10A**, **10B**, **10C** are meant, they are referred to generically as disc tools **10**.

- (24) The disc tools **10** each have a tool mount **11** which can be fastened to the output **85**. The tool mount **11** can, for example, be a bayonet tool mount with bayonet contours for fastening the output **85**. It is also possible that the tool mount **11** has, for example, a screw mount or passage opening for a fastening screw arranged on the output **85**. In any case, the disc tools **10** can be detachably fastened to the output **85** using the tool mounts **11**.
- (25) The tool mounts **11** are arranged on a machine side MS of the disc tools **10** provided for attachment to the processing machine **80**. A processing side BS, which is opposite the machine side MS, is used to process the workpiece W.
- (26) The tool mounts **11** are located in a center Z of the disc tools **10** through which an axis DA passes. The axis DA passes through the disc tools **10** vertically from the machine side MS to the processing side B. The axis DA can, for example, be an axis of rotation about which the disc tool can be driven in a rotary manner when a respective disc tool **10** is used on the processing machine **80**. The axis DA passes through a center Z of a respective disc tool **10**.
- (27) The disc tools **10** are, for example, essentially circular.
- (28) Furthermore, the disc tools **10** have flow-passage channels **12** that extend from the processing side BS to the machine side MS and have inflow openings **13** on the processing side BS, which are fluidically connected to outflow openings **14** on the machine side MS. The outflow openings **14** are arranged in the extraction housing **86** when the disc tool **10** is mounted on the processing machine **80**. Thus, dust-laden air SL can flow through a respective disc tool **10** in the direction of the machine side MS via the inflow openings **13**, where the dust-laden air SL flows into the extraction housing **86**. An extraction connection **87** communicates with the extraction housing **86**, to which, for example, an extraction hose of a vacuum cleaner can be connected.
- (29) The outflow openings **14** can be arranged on a cover **33** of the carrier **20** which provides or has the wall surface **22** in whole or in part and with which a base body **34** of the carrier **20** is covered on the machine side MS. The cover **33** covers, for example, the channel structures of the flow-passage channels **12** that can be seen in FIG. **10**, either completely or partially on the machine side MS.
- (30) The disc tools **10** have carriers **20**. The carriers **20** are, for example, plate-like. The carriers **20** have the same structure for all disc tools **10**, so that one carrier **20** is explained in the following description.
- (31) The carrier **20** has a mounting surface **21**, which is assigned to the processing side BS, and a wall surface **22** opposite the mounting surface **21**, which is at least partially covered by the extraction housing **86** when the disc tool **10** is arranged on the processing machine **80**. The outflow openings **14** are arranged in a ring around the center Z on a radially inner part of the wall surface **22**.
- (32) Inflow openings **23** are provided on the mounting surface **21**, through which inflow openings dust-laden air SL can flow into the carrier **20** and to the outflow openings **14**. The inflow openings **23** are arranged in a ring around a center Z or in a ring around the tool mount **11**. Furthermore, the inflow openings **23** extend radially inwards in the direction of the center Z of the carrier **20** or in the direction of the tool mount **11**, so that dust-laden air SL can also be extracted radially inwards with respect to an outer circumference **24** of the carrier **20**.
- (33) It is then possible in principle for the carrier **20** to have the tool mount **11** integrally. In the present case, however, there is a multi-part design, i.e. the carrier **20** has a mount **25** for a tool-mounting element **30** on which the tool mount **11** is arranged. The mount **25** has, for example, an insertion opening **26** that extends from the machine side to the processing side BS and from which the retaining mounts **27** extend radially outward like fingers. A base body **31** of the tool-mounting element **30** is accommodated in the insertion opening **26** and rests, for example, on the base **26A** thereof or opposite the base **26A**. Fastening arms **32** protrude radially outwards from the base body **31** with respect to the axis DA and engage the retaining mounts **27** in a form-fitting manner. The fastening arms **32** are bolted to the carrier **20** using screws **32A**.

(34) Retaining mounts **28** in the form of screw mounts **29** are provided on the mounting surface **21**, with which screw mounts the further components of disc tools **10A**, **10B**, and **10C** explained below can be attached or are attached to the respective carrier **20**. Each screw mount **29** has a screw thread **29A**, for example. Bolt sections **36** of retaining bolts **35** with the screw threads **36A** thereof can be screwed into the screw mount **29** or unscrewed from the screw mount **29**. For example, actuation contours **38**, for example slots, cross-slots, or the like, are provided on the bolt heads **37**, which slots can be actuated using an assembly tool, in particular a screwdriver. Thus, the cushion elements **40A**, **40B**, **40C** can be releasably attached to the carrier **20**.

(35) Bolt heads **37** of the retaining bolts **35** then support the further components of the disc tools **10** on the mounting surface **21** as explained below, in particular in terms of tensioning these further components with the mounting surface **21**, so that they are firmly held on the respective carrier **20**.

(36) The disc tools **10A**, **10B**, **10C** have cushion elements **40A**, **40B**, **40C** on the processing sides BS thereof.

(37) Each cushion element **40A**, **40B**, **40C** is sandwiched between a retaining wall element **41A**, **41B**, **41C** and a connecting wall element **42**, for example. Thus, the cushion elements **40A**, **40B**, **40C** can also be referred to as sandwich cushion elements.

(38) The cushion elements **40A**, **40B**, **40C** consist, for example, of rubber, foam, or other similar elastic material. For the cushion elements **40A**, **40B**, **40C**, the retaining wall elements **41A**, **41B**, **41C** and the connecting wall elements **42** are, so to speak, laminations or covers which ensure a uniform introduction of force onto a respective cushion element **40A**, **40B**, **40C**.

(39) For example, the connecting wall elements **42** and the retaining wall elements **41A**, **41B**, **41C** consist of a textile material, a fabric, in particular a textile fabric, a knitted fabric, a nonwoven, velour, or the like.

(40) In any case, at least the retaining wall element **41A**, **41B**, **41C** consists of a material that has a higher tensile strength than the cushion element **40A**, **40B**, **40C**.

(41) The connecting wall element **41A**, **41B**, **41C** preferably also consists of a material that has a higher tensile strength than the cushion elements **40A**, **40B**, **40C**.

(42) It is also possible for the retaining wall element **41A**, **41B**, **41C** and/or the connecting wall element **42** to consist of a VELCRO™ (hook and loop fastener) fabric which has VELCRO™ hooks.

(43) The cushion elements **40A**, **40B**, **40C** are firmly connected to the retaining wall elements **41A**, **41B**, **41C** and the connecting wall elements **42**, so that composite elements **43A**, **43B**, **43C** are formed. For example, the cushion elements **40A**, **40B**, **40C**, the retaining wall elements **41A**, **41B**, **41C**, and the connecting wall elements **42** are welded, glued, or similarly connected to one another in a material connection. A form-fitting connection, which can be provided in addition to the material-fitting connection or as an alternative to the material-fitting connection, is easily possible between the cushion element **40A**, **40B**, **40C** and the retaining wall element **41A**, **41B**, **41C** and the connecting wall element **42** of a respective composite element **43A**, **43B**, **43C**.

(44) The composite elements **43A**, **43B**, **43C** are attached to the carriers **20** as a whole by means of the retaining bolts **35**. The retaining bolts **35**, with the bolt sections **36** thereof, penetrate retaining bolt passage openings **48** which are flush with the retaining mounts **28** of the carrier **20** when a respective composite element **43A**, **43B**, **43C** is attached to the carrier **20**. Consequently, retaining bolt passage openings **48** are arranged in a ring around the axis DA or a center Z of a respective cushion element **40A**, **40B**, **40C**.

(45) The composite elements **43A**, **43B**, **43C** and thus the components thereof, namely the cushion element **40A**, **40B**, **40C**, the retaining wall element **41A**, **41B**, **41C**, and the connecting wall element **42**, have flow-passage openings or flow-passage channels **45**, which form components of flow-passage channels **12**. The flow-passage openings or flow-passage channels **45** are flush with the inflow openings **23** on the wall surface **22** of the carrier **20**. For example, flow-passage openings or flow-passage channels **45** arranged radially with respect to the axis DA and/or flow-

passage openings or flow-passage channels **45** arranged annularly along an outer circumference **44** are provided.

(46) The retaining wall elements **41A**, **41B**, **41C** serve to support an adhesive layer **50** to which a processing means **70** can be attached. The processing means **70**, for example, is an abrasive sheet **71**, which can be attached to the adhesive layer **50** with a mounting side **72** and has a processing side **73**, which is opposite the mounting side **72** and has a material suitable for processing the workpiece **W**, for example a grain, a nonwoven, or the like. The mounting side **72** has, for example, a loop fabric which is suitable for producing a VELCRO™ (hook and loop fastener) connection with VELCRO™ hooks **51** of the adhesive layer **50**.

(47) The processing means **70**, for example the abrasive sheets **71**, have flow-passage openings or flow-passage channels **75** that correspond to the inflow openings **13** or the flow-passage openings or the flow-passage channels **45** and are flush thereto when mounted on the disc tool **10**, so that dust-laden air can flow from the processing side **73** to the mounting side **72** and into the inflow openings **13**.

(48) In the case of disc tool **10B**, the adhesive layer **50** is arranged directly on the retaining wall element **41B**. Consequently, the retaining wall element **41B** forms an adhesive layer element **60B**.

(49) In the case of disc tools **10A** and **10C**, adhesive layer elements **60A**, **60C** are provided which are separate from the retaining wall elements **41A**, **41C** and have the adhesive layer **50**.

(50) The adhesive layer element **60A** has an adhesive layer cushion element **61** sandwiched between an intermediate wall element **62** and an adhesive layer wall **63A**. The adhesive layer cushion element **61**, together with the adhesive layer wall **63A** and the intermediate wall element **62**, forms an adhesive layer composite element **64**; the components thereof which are the adhesive layer wall **63A**, the cushion element **61**, and the intermediate wall element **62** are firmly connected to one another, for example glued together.

(51) The adhesive layer cushion element **61** is made of an elastic, resilient material such as foam, rubber, or the like.

(52) Similar to the composite element **43A**, **43B**, **43C**, the adhesive layer composite element **64** has the adhesive layer cushion element **61** reinforced or laminated by the outer components of the adhesive layer composite element **64**, namely the intermediate wall element **62** and the adhesive layer wall **63A**, because the intermediate wall element **62** and the adhesive layer wall **63A** consist, for example, of a material that has a higher tensile strength than the material of the adhesive layer cushion element **61**, for example a woven fabric, a knitted fabric, a nonwoven, or the like.

(53) The adhesive layer **50** is arranged on the adhesive layer wall **63A**. The adhesive layer **50** also has VELCRO™ hooks **51** in the adhesive layer element **60A**, which, however, are not shown in the drawing for reasons of simplification.

(54) The adhesive layer element **60C** consists only of an adhesive layer wall **63C**, which is arranged directly on the retaining wall element **41C**, for example glued thereto. The adhesive layer wall **63C** supports the adhesive layer **50** or has the adhesive layer **50**, i.e. it is provided with the VELCRO™ hooks **51**, for example (which is not shown in the drawing).

(55) Adhesive layer elements **60A** and **60C** are firmly connected to the retaining wall element **41A**, **41C**, for example glued and/or form-fittingly connected.

(56) For example, a combination of form-fitting and material-fitting connection is advantageous. For example, the intermediate wall element **62** and the retaining wall element **41A** can be designed to produce a VELCRO™ (hook and loop fastener) connection, wherein, for example, a VELCRO™ hooks are present on the retaining wall element **41A** and associated loops are on the intermediate wall element **62** or vice versa. Additional gluing, welding, or the like of the intermediate wall element **62** and the retaining wall element **41A** ensures a material-fitting, firm, and thus shear-resistant connection between the two composite elements **43A** and **64**, transversely to axis **DA**.

(57) The adhesive layer element **60C** has, for example, a VELCRO™ (hook and loop fastener)

layer or VELCRO™ hooks in the manner of the VELCRO™ hooks **51** according to FIG. **13**, wherein this not shown for reasons of simplification in the case of adhesive layer element **60C**.
(58) The retaining wall elements **41A**, **41B**, **41C** can be loaded by the bolt heads **37** when the retaining bolts **35** are screwed into the retaining mounts **28** in such a way that they tension and compress the cushion elements **40A**, **40B**, **40C** in the direction of the carrier **20**. The retaining wall elements **41A**, **41B**, **41C** form funnel-shaped or conical wall sections **141** next to the bolt heads **37**, which wall sections extend annularly around the bolt longitudinal axes BL of the retaining bolts **35** and/or the retaining bolt passage openings **48**. Close to the respective bolt head **37**, thus the material of the cushion element **40A**, **40B**, **40C** is almost completely displaced in the direction of the carrier **20** or the mounting surface **21**, while the force exerted onto the retaining wall element **41A**, **41B**, **41C** and thus onto the cushion element **40A**, **40B**, **40C** by the bolt head **37** is less at a greater radial distance from the bolt longitudinal axis BL of a respective retaining bolt **35** or the retaining bolt passage openings **48**.

(59) FIGS. **11**, **12** show a situation which arises without the measures, which are still to be explained. In addition to the funnel-shaped section **141**, material of the cushion element **40B** displaced by the retaining wall element **41B** and/or the retaining wall element **41B** itself forms a bulge **142**, which is generated by the application of force by the bolt head **37**. This bulge **142** also forms a bulge **143** on the adhesive layer **50** or on the retaining wall element **41B** so that the adhesive layer **50** also has an unevenness. Both the adhesive layer **50** and the processing means **70** or abrasive sheet **71** arranged thereon are thus subjected to greater pressure in the direction of the workpiece W by the bulge **143**, so that greater abrasion or more wear takes place, which can be seen in FIG. **11**.

(60) It is advantageously provided that the retaining wall element **41**, as shown by way of example on retaining wall element **41A**, has slots **49** which extend radially outwards and/or in a star shape around the retaining bolt passage openings **48**. The retaining wall element **41** adapts to the contour of the bolt heads **37** through the slots **49**.

(61) Such a bulge is avoided by the measures explained below, or the effect thereof is at least reduced in such a way that the adhesive layer **50** does not have a contour protruding towards the processing means **70** in the region and/or the vicinity of the bolt heads **37** of the retaining bolts **35**, but at most a depression is present, so that the unevenness or ring-shaped characteristics of the adhesive layer **50** shown in FIG. **11** are avoided.

(62) For this purpose, a recess **52** is provided in the adhesive layer **50** in the region and/or in the vicinity of the bolt heads **37**.

(63) The recesses **52** have cross-sections Q**52** which are larger than the cross-sections Q**48** of the retaining bolt passage openings **48**. The cross-sections Q**52** are provided for mounting the bolt heads **37**. A cross-section Q**52** is at least large enough for the bolt head **37** to be fully accommodated in the respective recess **52**. It is also possible, for example according to FIG. **6**, that a cross-section Q**52A** is provided for a recess **52**, which cross-section is greater than the cross-section of the bolt head **37** accommodated in the recess **52**.

(64) The bolt sections **36** of the retaining bolts **35** pass through retaining bolt passage openings **53** of the adhesive layer **50**. The recesses **52** extend around the retaining bolt passage openings **53**. As will become clear, the retaining bolt passage openings **53** can have a smaller inner cross-sectional area than the recesses **52**. However, it is also possible for the recesses **52** to have an inner cross-sectional area in which a respective bolt head **37** can be completely accommodated, so that the recess **52** at the same time forms a retaining bolt passage opening **53** for the bolt section **36** of a retaining bolt **35**.

(65) In the case of adhesive layer element **60A**, it is provided that each layer, namely the adhesive layer **50**, the adhesive layer wall **63A**, the adhesive layer cushion element **61**, and the intermediate wall element **62**, have passage openings **66**, **67**, and **68**. Passage openings **66**, **67**, and **68** advantageously have a larger cross-section and/or diameter than a depression **144** which results

from the tensioning of the retaining wall element **41A** and thus the compression of the cushion element **40A** by the bolt heads **37**. It is also possible for passage openings **66**, **67**, and **68** to have a cross-section and/or diameter that is at least as large as that of the depression **144**.

(66) The depressions **144** are, for example, funnel-shaped.

(67) If a bulge **145** in the manner of bulge **142** or **143** forms on the circumferential edge of a respective depression **144**, which is shown schematically in the drawing, this bulge **145** can be on the inner circumference of the recess **52**, for example on the inner circumference of the layers of the adhesive layer composite element **64** that are spaced apart from the adhesive layer **50**, for example on the inner circumference of passage openings **68** and/or **67**. Thus, passage openings **66**, **67**, and **68** and in any case the recess **52** form an expansion cavity or displacement cavity **69** for the material of the cushion element **40A**.

(68) However, the depressions **144** shown in FIGS. 2 and **10** do not result until the retaining bolts **35** are screwed into the carrier **20**. To illustrate this effect, however, the depressions **144** are illustrated in the composite element **43A** shown in perspective, even if the retaining bolts **35** are not shown.

(69) In the case of disc tool **10C**, the adhesive layer element **60C** has the adhesive layer **50**. In order to provide the recess **52**, the adhesive layer element **60C** has a passage opening **68**, the diameter of which is at least the same size as the depression **144** already explained, which results from the application of force onto the cushion element **40C** and the retaining wall element **41C** by a respective retaining bolt **35**. It is also possible for the diameter of the passage opening **68** to be larger than the diameter of the depression **144**. Any bulge **145** that may result from this can swell into the region or the plane of the adhesive layer **50** without protruding in front of the adhesive layer **50** in the direction of the processing means **70**. The recess or passage opening **68** on the adhesive layer element **60C** and therefore the adhesive layer **50** thereof thus forms a displacement cavity **69** for the material of the adhesive layer element **41C** and/or the cushion element **40C** and/or the composite element **43C**. However, due to the weakening and/or removal of the adhesive layer **50**, by means of which the recess **52** is formed, the respective bolt head **37** only rests on the retaining wall element **41C** and not on the adhesive layer **50**, so that the bulge **145** does not form or only to a small extent and/or the bulge **145** does not affect the geometry and/or the profile of the adhesive layer **50**.

(70) On the other hand, in the case of disc tool **10B**, the adhesive layer **50** is arranged integrally on the retaining wall element **41B**. However, the adhesive layer **50** is partially reduced in height or thickness or material strength, for example by means of a thermal and/or plastically deforming and/or pressure-loading processing tool **200** according to FIG. 14 or an abrasive or cutting processing tool **300** according to FIG. 15.

(71) The recesses **52** are formed as depressions in the adhesive layer **50** and/or the retaining wall element **41B**, wherein these depressions comprise a section **167** close to the retaining bolt passage openings **48** and a section **168** which has a larger radial distance to the bolt longitudinal axis BL. Consequently, the recesses **52** in the region of sections **167** are deeper in relation to the adhesive layer **50** or the free surface thereof for attaching the processing means **70**, while sections **168** are provided at a greater radial distance from the bolt longitudinal axis BL or the retaining bolt passage opening **48**, which sections are less deep behind the surface of the adhesive layer **50**.

(72) In the region of the sections **167**, the adhesive layer **50** and/or the retaining wall element **41B** is significantly weakened or no longer exists at all. This makes it possible, for example, for the load on the cushion element **40B** and/or the retaining wall element **41B** and/or the composite element **43B** generated by the bolt heads **37** of the retaining bolts **35** to be reduced and/or balanced, in particular with regard to the adhesive layer **50**. In this context, it can advantageously result that the displacement of the material of the cushion element **40B** and/or the composite element **43B** and/or the retaining wall element **41B** is reduced in terms of the formation of a bulge. However, it is also possible that a respective recess **52** accommodates a bulge that forms or is formed by a load on the

cushion element **40B** and/or the retaining wall element **41B** and/or the composite element **43B**.

(73) For example, the tool **200** comprises a support element **201**, which is opposite an embossing element **202**. On its side opposite the support element **201**, the embossed element **202** has an embossed contour **203** which interacts with a counter-bearing contour **204** of the support element **201**. Embossed contour **203** has a cross-section and/or diameter that corresponds to the cross-section or diameter of the recess **52** of the support element **10B** to be produced, i.e., for example, it has section **267** for producing section **167** of the depression or recess **52** and section **268** for producing section **168** of the recess **52**. The counter-bearing contour **204** has a profile that matches the embossed contour **203**. For example, the counter-bearing contour **204** and the embossing contour **203** have the same cross-sections or diameters.

(74) The support element **201** and the embossed element **202** can have centering contours **205** and **206** for centering with respect to one another. Centering contour **205** is designed as a mount, for example, while centering contour **206** is designed as a projection that can be accommodated by the mount or centering contour **205**. The centering projection or the centering contour **206**, for example, can penetrate a respective retaining bolt passage opening **48** and then penetrate the mount or centering contour **205**.

(75) The embossing element **202** can be adjusted between an embossing position or processing position WP illustrated in dashed lines and a rest position RP illustrated in solid lines by a schematically illustrated actuator **207**.

(76) Furthermore, the embossing element **202** is heated using a heater **208**. If the embossing element **202** acts on the adhesive layer **50** of the composite element **43B** in the manner of an embossing die, the VELCRO™ hooks **51** deform to different extents, which is shown in FIG. **13** as an example for VELCRO™ hooks **51A** and **51B**. In addition to the depression **52**, the VELCRO™ hooks **51** project completely in front of the retaining wall element **41B**, while they project less far in front of the retaining wall element **41B** in the region of the depression **52**, that is to say they have completely or partially melted away. In principle, it is possible for the VELCRO™ hooks **51** to at least partially melt due to the load from the embossing element **202** and the thermal effect.

(77) At this point it should be mentioned that embossing element **202** could also be operated without heating, so that only a mechanical, plastic deformation of the adhesive layer **50** is effected.

(78) Alternatively, it is also possible that embossing element **202** does not have the function of an embossing element, but only provides a deformation contour that acts on the adhesive layer **50** with little force due to the heater **208**, so that a thermal deformation of the adhesive layer **50** can essentially be achieved.

(79) A grinding or abrasive processing is shown in FIG. **15** in connection with the processing tool **300**. The processing tool **300** has a support element **301** in the manner of support element **201**, which is opposite a grinding tool **302** in the manner of the embossing stamp or embossing element **202**.

(80) A counter-bearing contour **304** in the manner of counter-bearing contour **204** is arranged on the support element **301**. Support element **301** also has a centering contour **305** in the manner of centering contour **205**, which interacts with a centering contour **306** of the processing element **302**, i.e., for example, in the manner in which centering contour **206** is designed as a centering projection that can penetrate into a retaining bolt passage opening **48** and into the centering contour **205** designed as a centering mount.

(81) The processing element **302** comprises a support body **310** which can be adjusted, by an actuator **307** in the manner of actuator **207**, between a processing position WP shown in dashed lines and a rest position RP shown in solid lines.

(82) A bearing mount **311** for a bearing **321** of a grinding tool **320**, which is mounted on the support body **310** so as to be rotatable about an axis of rotation RO, is arranged on the support body **310**. The grinding tool **320** can be driven in rotation about the axis of rotation RO by a rotary drive **322**, so that a grinding contour **303**, which is opposite the support element **301** or the counter-

bearing contour **304**, can grind the depression **52** into the retaining wall element **41B**. Like the embossing contour **203**, the grinding contour **303** is adapted to the contour of the recess **52** to be produced.

(83) An exemplary embodiment shown in FIG. **16** as a variant of the exemplary embodiment in FIG. **3** of a disc tool **10D** can provide, for example, that a retaining wall element **41D** in the manner of retaining wall element **41A** is present in order to retain the adhesive layer cushion element **61** on the carrier **20**. In contrast to the exemplary embodiment according to FIG. **3**, however, it is provided that the retaining wall element **41D** is screwed directly onto the carrier **20**, i.e., in contrast to the exemplary embodiment according to FIG. **3**, there is no cushion element **40A** and no connecting wall element **42**. However, it is also possible for the retaining wall element **41D** and the connecting wall element **42** (not shown) to be present, but not the cushion element **40A**.

10 Disc tool **ABC** **60** Adhesive layer element **ABC** **11** Tool mount **61** Adhesive layer cushion element **12** Flow-passage channels **62** Intermediate wall element **13** Inflow openings **63** Adhesive layer wall A, C **14** Outflow openings **64** Adhesive layer composite element **15** **65** Flow-passage channels for **12** **16** **66** Passage opening **17** **67** Passage opening **18** **68** Passage opening **19** **69** Displacement cavity **20** Carrier **70** Processing means **21** Mounting surface **71** Abrasive sheet **22** Wall surface **72** Mounting side **23** Inflow openings **73** Processing side **24** Outer circumference **74** **25** Mount for **30** **75** Flow-passage channels for **12** **26** Insertion opening base **26A** **76** **27** Retaining mounts for **32** **77** **28** Retaining mounts for **35** **78** **29** Screw mounts screw thread **29A** **79** **30** Tool-mounting element **80** Processing machine **31** Base body **81** Housing **32** Fastening arms **82** Handle **33** Cover **83** Drive motor **34** Base body **84** Gear **35** Retaining bolts **85** Output **36** Bolt section screw thread **36A** **86** Extraction housing **37** Bolt heads **87** Extraction connection **38** Actuating contour slots **88** **40** Cushion elements **ABC** **90** **41** Retaining wall element **ABCD** **91** **42** Connecting wall element **92** **43** Composite element **ABC** **93** **44** Outer circumference **94** **45** Flow-passage channels for **12** **95** **46** **96** **47** **97** **48** Retaining bolt passage openings **98** **49** Slots **99** **50** Adhesive layer **W** Workpiece **51** VELCRO™ hook **MS** Machine side **52** Recesses **BS** Processing side **53** Retaining bolt passage openings **DA** Axis **54** **BL** Bolt longitudinal axis

Claims

1. A disc tool for a processing machine for grinding and/or polishing a workpiece, wherein the disc tool comprises: a carrier for attachment to an output of the processing machine, a cushion element made of elastic material arranged on the carrier; and a retaining wall element for retaining the cushion element arranged on the carrier, wherein the retaining wall element is retained on the carrier by means of retaining bolts, wherein respective bolt portions of the retaining bolts extending along respective bolt longitudinal axes and extending through respective retaining bolt passage openings of the retaining wall element are accommodated in respective retaining mounts of the carrier, and the retaining wall element is held between bolt heads of the retaining bolts and the carrier, and wherein an adhesive layer is arranged on a side of the retaining wall element opposite the carrier, and wherein a processing means for grinding and/or polishing the workpiece is releasably attached to the adhesive layer, and wherein the adhesive layer has recesses associated with the bolt heads, the recesses being present before the retaining bolts are screwed into the carrier and cross-sections of the recesses provided to mount the bolt heads are greater than cross-sections of the retaining bolt passage openings respectively, and wherein the disc tool further comprises an adhesive layer element which has an adhesive layer wall with the adhesive layer and is arranged on the retaining wall element, and wherein the retaining wall element has a hook arrangement and/or hooks for retaining the adhesive layer element.
2. The disc tool according to claim 1, wherein: a central axis or a center of at least one recess of the recesses is concentric with respect to the bolt longitudinal axis of the retaining bolt accommodated in the at least one recess of the recesses; and/or at least one of the recesses or all of the recesses of

the adhesive layer has or have an inner diameter that is at least as large as an outer diameter of the bolt head accommodated in the recess or is larger than the outer diameter of the bolt head accommodated in the recess; and/or a material thickness of the adhesive layer or of the adhesive layer element having and/or supporting the adhesive layer is reduced to provide the recesses; and/or the adhesive layer has passage openings through which the bolt heads can penetrate the adhesive layer; and/or at least one of the recesses or all of the recesses of the adhesive layer has or have a larger cross-section and/or diameter than the bolt heads, so that a distance is advantageously present between an inner circumference of a respective recess and an outer circumference of the bolt head accommodated in the recess, transverse to the bolt longitudinal axis thereof.

3. The disc tool according to claim 1, wherein at least one of the recesses or all of the recesses forms or form a displacement cavity into which components of the retaining wall element and/or of the cushion element can be displaced or are displaced.

4. The disc tool according to claim 1, wherein at least one of the recesses or all of the recesses is or are formed and/or designed for accommodating a bulge of the cushion element and/or of the retaining wall element.

5. The disc tool according to claim 1, wherein at least one of the recesses or all of the recesses is or are deeper with respect to the adhesive layer in a region radially closer to the bolt longitudinal axis of the retaining bolt accommodated therein than in a region further away radially from the bolt longitudinal axis.

6. The disc tool according to claim 1, wherein the retaining wall element consists of a material with a higher tensile strength than the cushion element.

7. The disc tool according to claim 1, wherein the cushion element is arranged as an intermediate layer between the carrier and the retaining wall element.

8. The disc tool according to claim 7, wherein the bolt portions of the retaining bolts pass through the retaining wall element and the cushion element, and/or the cushion element is compressed between the bolt heads of the retaining bolts and the carrier.

9. The disc tool according to claim 7, wherein a connecting wall element is arranged between the cushion element and the carrier, wherein the connecting wall element consists of a material with a higher tensile strength than the elastic material of the cushion element.

10. The disc tool according to claim 9, wherein the cushion element and the retaining wall element and/or the cushion element and the connecting wall element form a sandwich-like connected unit or a composite element, on which the recesses are formed.

11. The disc tool according to claim 1, wherein the recesses are formed by a thermal and/or machining and/or abrasive and/or cutting and/or pressure-loading and/or embossing process that reduces a material thickness of the adhesive the layer and/or of the adhesive layer element having the adhesive layer.

12. The disc tool according to claim 1, wherein the adhesive layer element is held on the retaining wall element in a materially-fitting and/or form-fitting manner.

13. The disc tool according to claim 1, wherein the adhesive layer has a layer with a hook and loop fastener arrangement for retaining the processing means.

14. The disc tool according to claim 1, wherein the adhesive layer and the cushion element consist of different materials and/or are different layers.

15. The disc tool according to claim 1, wherein the disc tool further comprises flow-passage openings and/or an opening for a tool mount of the disc tool provided for attachment to the output of the processing machine, and wherein the adhesive layer completely or essentially completely covers a side of the disc tool facing away from the carrier, apart from the flow-passage openings and/or the opening for a tool mount of the disc tool.

16. The disc tool according to claim 1, wherein slots extend radially away from the retaining bolt passage openings, so that a region of the retaining wall element surrounding a respective retaining bolt passage opening is segmented.

17. The disc tool according to claim 1, wherein a tool mount or a tool-mounting element having a tool mount is arranged on the carrier, wherein the disc tool can be detachably connected to the output of the processing machine by means of the tool mount.

18. The disc tool according to claim 1, further comprising at least one flow-passage channel for air, which has inflow openings on the adhesive layer and outflow openings on the carrier, wherein air can flow through the at least one flow-passage channel from the adhesive layer in the direction of the carrier or, vice versa, from the carrier in the direction of the adhesive layer.

19. The disc tool according to claim 1, wherein the processing means is an abrasive sheet and/or the disc tool comprises the processing means.

20. The processing machine with the disc tool according to claim 1, wherein the processing machine has a drive with a drive motor for the disc tool, by means of which the disc tool can be driven by the processing machine to perform a rotary and/or eccentric and/or oscillatory movement.

21. The processing machine according to claim 20, further comprising a long-neck grinder.

22. A disc tool for a processing machine for grinding and/or polishing a workpiece, wherein the disc tool comprises: a carrier for attachment to an output of the processing machine, a cushion element made of elastic material arranged on the carrier; and a retaining wall element for retaining the cushion element arranged on the carrier, wherein the retaining wall element is retained on the carrier by means of retaining bolts, wherein respective bolt portions of the retaining bolts extending along respective bolt longitudinal axes and extending through respective retaining bolt passage openings of the retaining wall element are accommodated in respective retaining mounts of the carrier, and the retaining wall element is held between bolt heads of the retaining bolts and the carrier, and wherein an adhesive layer is arranged on a side of the retaining wall element opposite the carrier, and wherein a processing means for grinding and/or polishing the workpiece is releasably attached to the adhesive layer, and wherein the adhesive layer has recesses associated with the bolt heads, the recesses being present before the retaining bolts are screwed into the carrier and cross-sections of the recesses provided to mount the bolt heads are greater than cross-sections of the retaining bolt passage openings respectively, and wherein the disc tool further comprises an adhesive layer element which has an adhesive layer wall with the adhesive layer and is arranged on the retaining wall element, and wherein the adhesive layer element has at least one adhesive layer cushion element made of an elastic material arranged between the adhesive layer wall and the retaining wall element.

23. The disc tool according to claim 22, wherein the retaining wall element has a hook arrangement and/or hooks for retaining the adhesive layer element.

24. The disc tool according to claim 22, wherein the at least one adhesive layer cushion element has passage openings through which the bolt heads penetrate and/or through which the bolt heads can be inserted or have been inserted freely, and/or the at least one adhesive layer cushion element is not tensioned by the bolt heads against the carrier and/or the at least one adhesive layer cushion element is arranged on the adhesive layer element on a side facing away from the bolt heads.

25. The disc tool according to claim 22, wherein the adhesive layer element has an intermediate wall element which is arranged between the adhesive layer cushion element and the retaining wall element and is connected to the retaining wall element.

26. The disc tool according to claim 25, wherein the intermediate wall element consists of a material which has a higher tensile strength than the adhesive layer cushion element.

27. A disc tool for a processing machine for grinding and/or polishing a workpiece, wherein the disc tool comprises: a carrier for attachment to an output of the processing machine, a cushion element made of elastic material arranged on the carrier; and a retaining wall element for retaining the cushion element arranged on the carrier, wherein the retaining wall element is retained on the carrier by means of retaining bolts, wherein respective bolt portions of the retaining bolts extending along respective bolt longitudinal axes and extending through respective retaining bolt passage openings of the retaining wall element are accommodated in respective retaining mounts of the

carrier, and the retaining wall element is held between bolt heads of the retaining bolts and the carrier, and wherein an adhesive layer is arranged on a side of the retaining wall element opposite the carrier, and wherein a processing means for grinding and/or polishing the workpiece is releasably attached to the adhesive layer, and wherein the adhesive layer has recesses associated with the bolt heads, the recesses being present before the retaining bolts are screwed into the carrier and cross-sections of the recesses provided to mount the bolt heads are greater than cross-sections of the retaining bolt passage openings respectively, and wherein the disc tool further comprises an adhesive layer element which has an adhesive layer wall with the adhesive layer and is arranged on the retaining wall element, and wherein the adhesive layer element has at least one adhesive layer cushion element made of an elastic material arranged between the adhesive layer wall and the retaining wall element, and wherein the adhesive layer element has an intermediate wall element which is arranged between the adhesive layer cushion element and the retaining wall element and is connected to the retaining wall element, and wherein the adhesive layer cushion element and the intermediate wall element and/or the adhesive layer wall form a sandwich-like connected unit of the adhesive layer element or an adhesive layer composite element of the adhesive layer element, on which the recesses are formed.
